The Cellular Coding of Temperature in the Mammalian Cortex

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Temperature is a fundamental sensory modality separate from touch, with dedicated receptor channels and primary afferent neurons for cool and warm. Unlike for other modalities, however, the cortical encoding of temperature remains unknown, with very few cortical neurons reported that respond to non-painful temperature, and the presence of a 'thermal cortex' is debated. Here, using widefield and two-photon calcium imaging in the mouse forepaw system, we identify cortical neurons that respond to cooling and/or warming with distinct spatial and temporal response properties. We observed a representation of cool, but not warm, in the primary somatosensory cortex, but cool and warm in the posterior insular cortex (pIC). The representation of thermal information in pIC is robust and somatotopically arranged, and reversible manipulations show a profound impact on thermal perception. Despite being positioned along the same one-dimensional sensory axis, the encoding of cool and that of warm are distinct, both in highly and broadly tuned neurons.

Experimental Setup and pIC Somatotopic Map

A fundamental question in neuroscience is how the external sensory environment is represented in the cortex. In the thermal system, at present there is no consensus on how or where sensory information is encoded. One model is that cool and warm are processed by functionally and anatomically segregated circuits, following labelled-line principles. Another is that cool and warm are integrated in the thermal pathway resulting in cortical neurons with a continuous, graded representation of temperature.

To examine thermal processing in S1 and pIC, we carried out widefield calcium imaging through a glass window in paw-tethered, awake mice expressing the calcium indicator GCaMP6s in cortical excitatory neurons. We presented an 8-kHz acoustic stimulus to locate the anterior auditory field in the auditory cortex and the small auditory insular field that border the somatosensory regions of pIC.

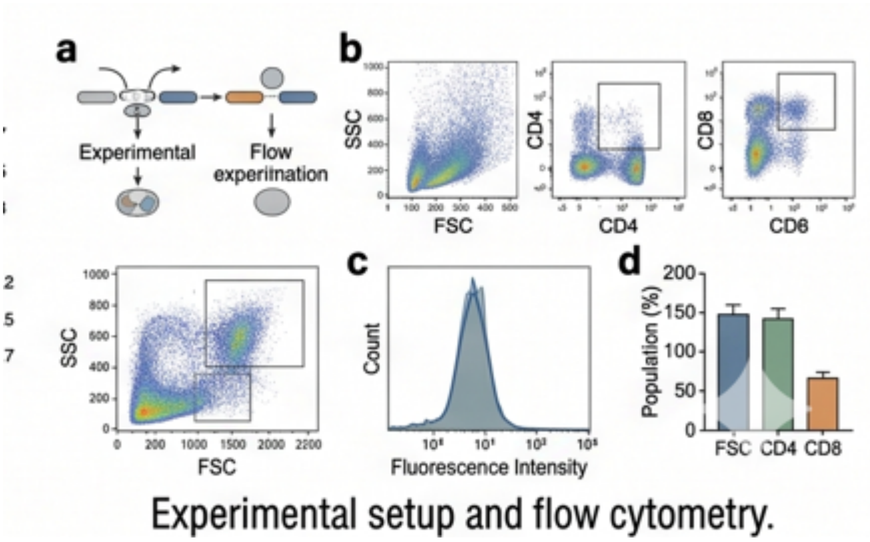

Figure 1: Experimental setup and flow cytometry. (a) Schematic showing the experimental setup and flow experimentation process. (b) Flow cytometry scatter plots detailing Side Scatter (SSC) vs. Forward Scatter (FSC), alongside gating for CD4 and CD8 cell populations. (c) Histogram detailing count versus fluorescence intensity. (d) Population percentages for FSC, CD4, and CD8.

Widefield calcium imaging of pIC showed a clear somatotopic arrangement of thermal and tactile responses anterior to the auditory cortex in all mice, with the face represented in a region anterior and ventral to the forepaw, which in turn is anterior to the hindpaw. In all mice, the forepaw had a larger-amplitude response to thermal stimuli than the hindpaw, perhaps reflecting the dominant role of the forepaw in haptic exploration.

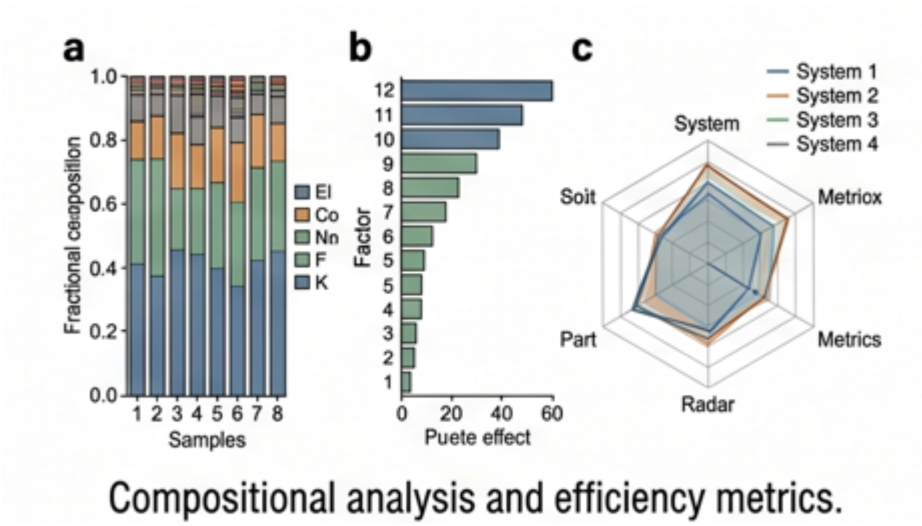

Figure 2: Compositional analysis and efficiency metrics. (a) Stacked bar chart showing the fractional composition of elements across 8 distinct samples. (b) Horizontal bar chart ranking 12 different factors based on their effect size. (c) Radar chart comparing System 1 through 4.

Thermal Tuning and Topography of pIC Neurons

Cooling and warming widefield responses in pIC overlap spatially. This could result from an intermingled distribution of highly tuned or broadly tuned neurons. To determine the thermal tuning of cortical neurons, we went on to carry out two-photon calcium imaging of pIC excitatory neurons in awake mice and observed robust cellular responses to cooling and warming.

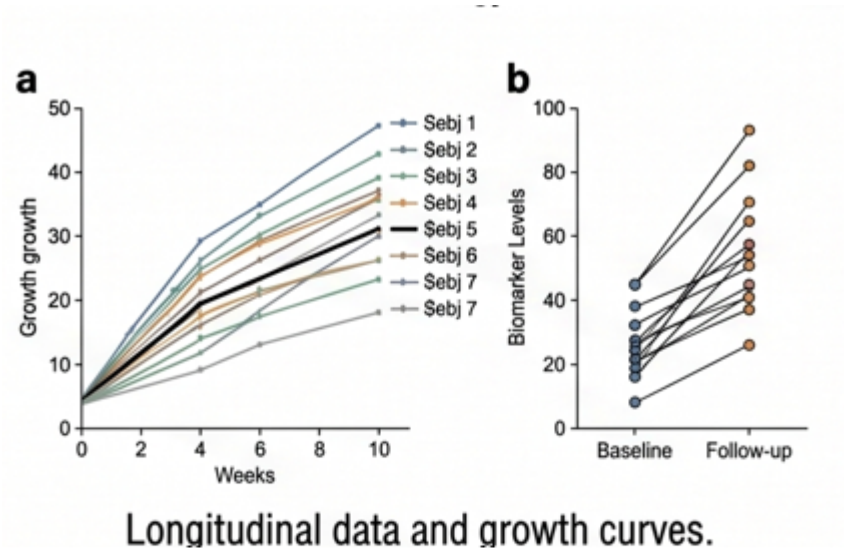

Figure 3: Longitudinal data and growth curves. (a) Line graphs tracking growth over a 10-week period for 7 subjects. (b) Paired scatter plot showing individual biomarker levels from Baseline to Follow-up.

We plotted the normalized responses according to a thermal bias index that describes the relative response strength to cooling compared to warming. The index had a U-shaped distribution, suggesting that segregated channels of tuned input drive cool and warm responses, and showed a similar probability of highly tuned neurons (cool only or warm only) and broadly tuned neurons (cool and warm responsive).

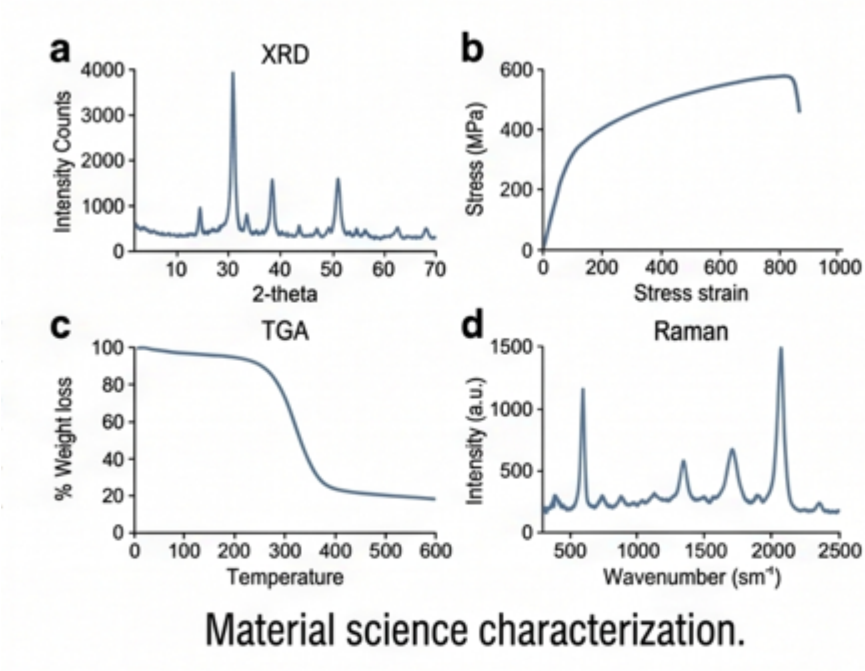

Figure 4: Material science characterization. (a) X-ray Diffraction (XRD) pattern. (b) Stress-strain curve demonstrating material elasticity up to nearly 600 MPa. (c) Thermogravimetric Analysis (TGA). (d) Raman spectroscopy plotting intensity against wavenumber.

Temporal Dynamics of Cortical Thermal Responses

The skin contains discrete spots that evoke cool or warm sensations. Notably, in vivo recordings from sensory afferent neurons, which innervate thermal spots, have shown distinct response dynamics to thermal stimuli, with short-latency, transient cool response mediated by Ad-fibres, and longer-latency, sustained firing in response to cool and warm stimuli by C-fibres.

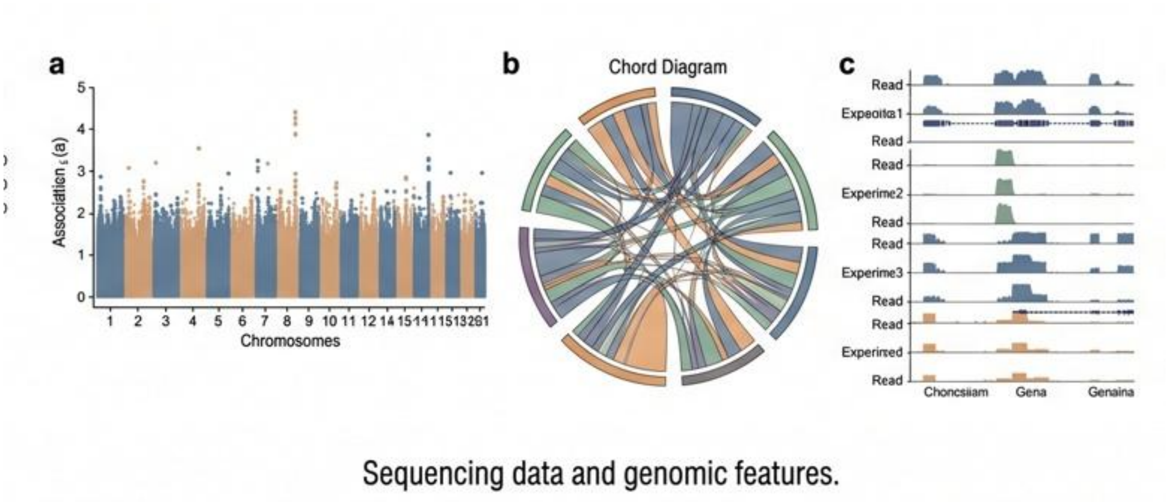

Figure 5: Sequencing data and genomic features. (a) Manhattan plot displaying genetic associations across 21 chromosomes. (b) Chord diagram illustrating complex genomic interactions and linkages.

In remarkable similarity to the case for the periphery, a grand average of all significant cool and warm responding neurons in pIC showed a shorter-latency (about 80 ms) and more transient cool response compared to a longer-latency (about 320 ms) and more sustained warm response. The bimodal distribution of the duration index for cool responses suggests a transient and a sustained response type, whereas pIC warm responses show a broad distribution of sustained responses.

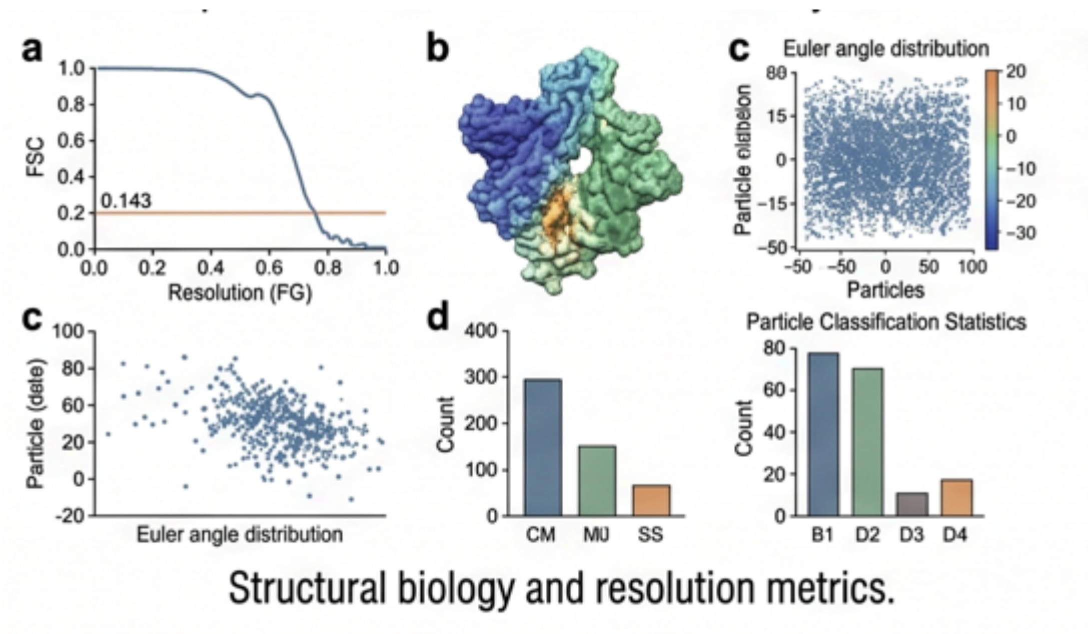

Figure 6: Structural biology and resolution metrics. (a) Fourier Shell Correlation (FSC) curve crossing the 0.143 resolution threshold. (b) 3D reconstruction of a protein complex.

Encoding of Cooling and Warming Amplitude

The amplitude of a sensory stimulus can be encoded in different ways: by neurons with specific preferred amplitude values; the recruiting of more neurons as amplitude increases; graded rate coding. To address this in the thermal system, we measured single-cell responses in pIC to different amplitude cooling and warming stimuli and show that responses to different amplitudes were reliable within a neuron, but diverse between neurons.

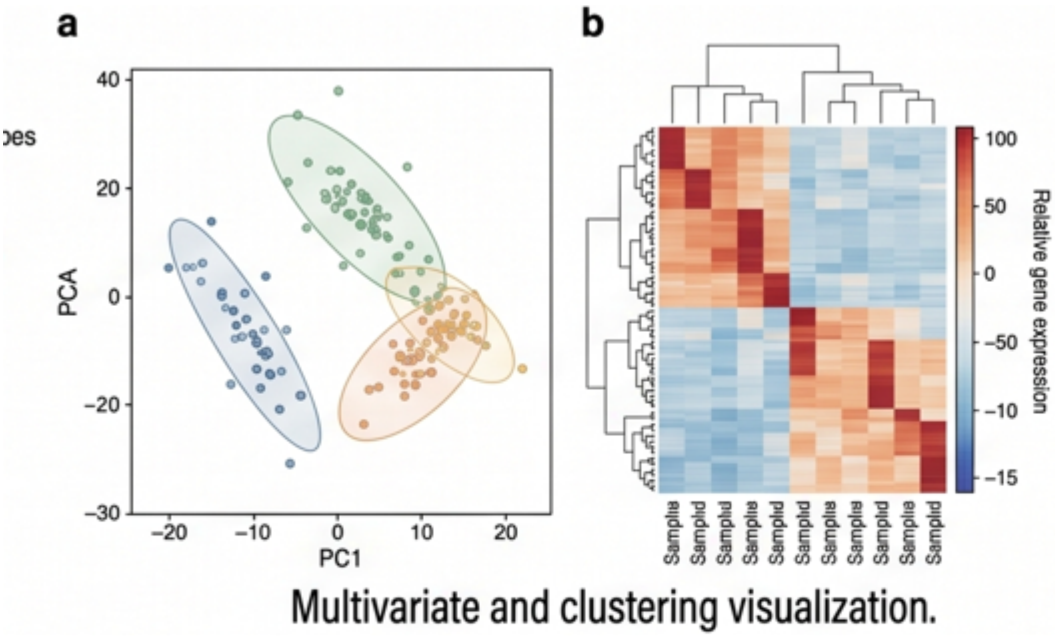

Figure 7: Multivariate and clustering visualization. (a) Principal Component Analysis (PCA) scatter plot revealing three distinct sample clusters. (b) Heatmap with dendrograms showing relative gene expression across multiple samples.

Plotting the population response amplitude showed an asymmetric response profile with a steeper curve for cooling and more graded for warming. In pIC, most cool neurons respond to the smallest amplitude tested with only a minority recruited at larger amplitudes, whereas the number of activated warm neurons more gradually follows the increase of stimulus amplitude.

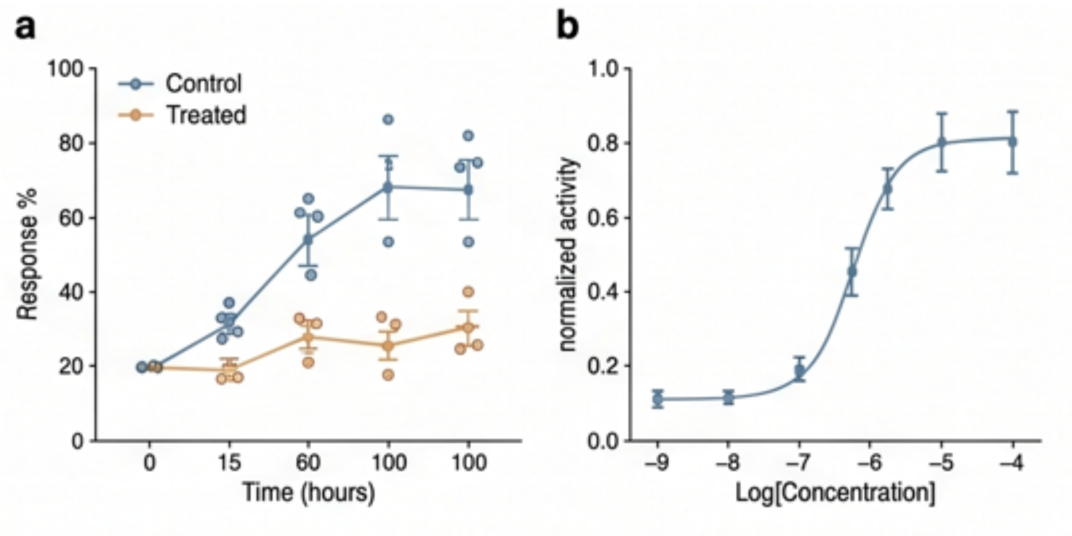

Figure 8: Pharmacological dose-response and kinetics. (a) Time-course graph comparing response % over 100 hours between Control and Treated groups. (b) Dose-response curve showing a standard sigmoidal relationship.

Thermal Perception is Mediated by pIC

To address the link between thermal sensitivity and perception, we carried out optogenetic manipulations during a go/no-go thermal detection task. VGAT-ChR2 mice were implanted with an optical window over pIC and provided with water rewards on licking a water spout during the onset ramp or plateau phase of the stimulus.

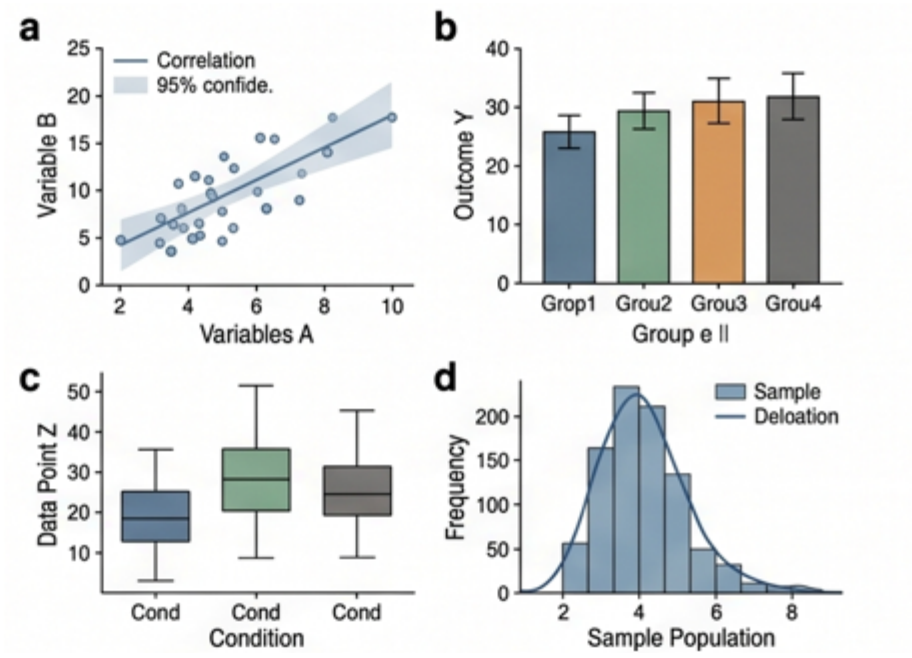

Figure 9: Statistical correlations and distributions. (a) Scatter plot showing a positive linear correlation between Variable A and Variable B. (b) Bar chart comparing Outcome Y across distinct experimental groups. (c) Box plots comparing Data Point Z. (d) Histogram with an overlaid density curve.

Mice were able to report stimuli at all amplitudes, highlighting their acute thermal perceptual ability. As for the functional responses, mice reported warming at longer latencies than for cooling. In cooling-trained mice, optogenetic inhibition of pIC suppressed the hit rates to 2°C, 1°C and 0.5°C stimuli, whereas in warming-trained mice, optogenetic inhibition suppressed the hit rates to all amplitudes.

Discussion and Abstract Visualizations

Our data show that humans, monkeys and rodents therefore share not only similar perceptual abilities, but also a cortical area (pIC) specialized in temperature processing, putting the thermosensory system of mammals closer than previously thought. The presence of distinct features of cool and warm responses both in broadly and tightly tuned pIC neurons shows that functionally segregated streams of afferent information can merge in cortical neurons.

Figure 10: Abstract representation of sensory integration. A metaphorical depiction of the "thermal cortex" blooming with sensory inputs, much like the vibrant array of flowers emerging from the astronaut's helmet, reflecting the complex and intertwined nature of thermosensory processing.

The discovery of an optically accessible cortical representation of temperature provides a platform to address the neural mechanisms of non-painful thermal perception, as well as pain evoked during noxious stimulation, allodynia and thermal illusions.